

| Question |     |      | Marking details                                                                                                                                                                                                                                                            | Marks available |          |          |          | Maths    | Prac     |
|----------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |                                                                                                                                                                                                                                                                            | AO1             | AO2      | AO3      | Total    |          |          |
| 6        | (a) | (i)  | 0.327 m (1)<br>± 1.5% (1)<br>No s.f. penalty                                                                                                                                                                                                                               |                 | 2        |          | 2        | 2        | 2        |
|          |     | (ii) | 335 m s <sup>-1</sup> (1) <b>ecf</b> [allow 334.9]<br>± 8 m s <sup>-1</sup> (1) <b>ecf</b> [allow 8.4]<br>Allow 1 <sub>max</sub> for incorrect / inconsistent sf                                                                                                           |                 | 2        |          | 2        | 2        | 2        |
|          | (b) | (i)  | $x = \frac{\lambda}{4}$ or $4x = \lambda(1)$<br>Use of $v = f\lambda$ (1)                                                                                                                                                                                                  |                 | 2        |          | 2        | 1        | 2        |
|          |     | (ii) | Node, antinode... sequence with antinode at top and node at bottom marked or equiv e.g. string of sausages diagram, with at least one node between top and bottom of air column(1)<br>Correct sequence [A, N, A, N] going downwards or equiv with spacings correct by eye. | 1               | 1        |          | 2        |          | 2        |
|          |     |      | <b>Question 6 total</b>                                                                                                                                                                                                                                                    | <b>1</b>        | <b>7</b> | <b>0</b> | <b>8</b> | <b>5</b> | <b>8</b> |